ISMAEL ORTIZ
Department of Bioengineering
Rice University, Houston, TX
Email: io11@rice.edu
ismaelortiz.bme@gmail.com
LinkedIn

EDUCATION

2024-Present	Rice University, Houston, TX PhD Candidate in Bioengineering Advisor: Cynthia A. Reinhart-King, PhD
2020-2024	Vanderbilt University, Nashville, TN PhD Student in Biomedical Engineering Advisor: Cynthia A. Reinhart-King, PhD
2016-2020	University of Miami, Coral Gables, FL BS in Biomedical Engineering, Pre-Med Concentration Documentary Ideas and Practice Cognate Foundations of Behavior for Pre-health Professions Cognate

RESEARCH EXPERIENCE

August 2020 – Present	Graduate Research Assistant – Reinhart-King Laboratory Department of Biomedical Engineering Advisor: Cynthia A. Reinhart-King PhD
September 2018 – May 2020	Undergraduate Research Assistant – Physiomimetic Microsystems Laboratory Department of Biomedical Engineering Advisor: Ashutosh Agarwal, PhD
March 2018 – January 2019	Undergraduate Research Assistant – Water Quality Engineering Laboratory Department of Environmental Engineering Advisor: James D. Englehardt, PhD
May 2017 – October 2017	Undergraduate Research Assistant – Physiologic Imaging and Modeling Lab Department of Radiology and Biomedical Engineering Advisor: Noam Alperin, PhD

AWARDS AND HONORS

2022 – 2025	National Science Foundation Graduate Research Fellowship
2022 – 2025	Ford Foundation Predoctoral Fellowship
2021	Richard Bennet/Dorothy Danforth Compton Prize Recognizes the best science/engineering research proposal from a first-year minority Vanderbilt University PhD student.
2020 – 2022	NIH Diversity Supplement Award
2020 – 2023	Provost's Graduate Fellow
2020 – 2023	Russel G. Hamilton Scholar
2020	Provost's Honor Roll
2019	Dean's List
2018	Dean's List
2016 – 2020	Canes Achievement Award
2016 – 2020	Florida Bright Future's Medallion Scholar

PUBLICATIONS

** Authors contributed equally*

Research Articles

1. **I Ortiz**, V.L. Dunagan, V Tupule, W Wang, G Vasiukov, M.R. King, C.A. Reinhart-King. Shear Activation of Ca^{2+} Ion Channels Mediates Mechanosensitive ATP Recovery in Cancer Cells (*in preparation to be submitted*)
2. **I Ortiz**, V.L. Dunagan, J.D. Greenlee, K.M. Young, J.V. Hernandez, P.V. Taufalele, M Lopez-Cavestany, W Wang, F Bordeleau, M.R. King, C.A. Reinhart-King. Fluid Shear Stress Enriches Mesenchymal Circulating Tumor Cells Through Mechanical Selection. *Journal of Cell Science*. October 2026. https://doi.org/10.1242/jcs.265092
3. **I. Ortiz*,** P. V. Taufalele*, V. L. Dunagan, L. A. Hapach, J. Wang, Q. Liu, C. A. Reinhart-King. Transcriptional heterogeneity underlying cancer cell migration converges on shared regulatory programs. *Genes*. August 2026. doi: https://doi.org/10.3390/genes17091042
4. M.R. Zanotelli, J.P. Miller, W. Wang, **I. Ortiz**, E. Tahon, F. Bordeleau, C.A. Reinhart-King. Tension directs cancer cell migration over fiber alignment through energy minimization. *Biomaterials*. December 2024. doi: 10.1016/j.biomaterials.2024.122682
5. S.M. King*, **I. Ortiz***, N. S. Sarna*, W. Wang, M. Lopez-Cavestany, Z. Zhang. Retention of E-selectin functionalized liposome fanny packs on Jurkat cells following invasion through collagen. *Journal of Immunological Methods*. August 2024. doi: 10.1016/j.jim.2024.113700
6. S.C. Schwager*, J.A. Mosier*, R.S. Padmanabhan, A. White, Q Xing, L.A. Hapach, P.V. Taufalele, **I. Ortiz**, C.A. Reinhart-King. Link between glucose metabolism and EMT drives triple negative breast cancer migratory heterogeneity. *iScience*. October 2022. doi: 10.1016/j.isci.2022.105190
7. M.R. Zanotelli, J. Zhang, **I. Ortiz**, W. Wang, N. Chada, C.A. Reinhart-King. Highly motile cells are metabolically responsive to collagen density. *Proceedings of the National Academy of Sciences*. April 2022. doi: 10.1073/pnas.2114672119
8. V. Mayo, A. Bowles, L. Wubker, **I. Ortiz**, A. Cordoves, R. Cote, D. Correa, A. Agarwal. Human-derived osteoblast-like cells and pericyte-like cells induce distinct metastatic phenotypes in primary breast cancer cells. *Experimental Biology and Medicine*. November 2020. doi: 10.1177/1535370220971599

Review Articles

1. **I. Ortiz**, S. Lopez, R. Kondapaneni, J. V. Hernandez. Cellular mechanometabolism: stimuli and implications. *EMBO Reports*. May 2026. https://doi.org/10.1038/s44319-026-00795-4

PRESENTATIONS

1. **I. Ortiz**, V. L. Dunagan, N. S. Sarna, W Wang, G Vasiukov, MR King, CA Reinhart-King. Cancer Cell Glucose Metabolic Reprogramming via A Fluid Shear Stress Activated PI3K Pathway. *Biomedical Engineering Society 2024 Annual Conference*, Baltimore MD 2024
2. **I. Ortiz**, J. D. Greenlee, W. Wang, K. M. Young, V. L. Dunagan, P. V. Taufalele, M. Lopez-Cavestany, M. R. King, C. A. Reinhart-King. Cytoskeletal Actin Organization of Migratory Cancer Cells Aids Single-Cell Fluid Shear Stress Resistance. *Biomedical Engineering Society 2023 Annual Conference*, Seattle WA, October 2023
3. P.V. Taufalele*, **I. Ortiz***, J. Wang, Q. Liu, C.A. Reinhart-King. Transcriptomic Analysis Suggests Biological Processes are Shared Among Highly Migratory Cancer Cells but Accomplished Through Unique Transcriptional Programs. *Biomedical Engineering Society 2023 Annual Conference*, Seattle WA, October 2023
4. V.L. Dunagan, **I. Ortiz**, C. A. Reinhart-King. Migration Based Sorting of Breast Cancer Cells Reveals the Bi-Directional Relationship with Epithelial to Mesenchymal Phenotypes. *Biomedical Engineering Society 2023 Annual Conference*, Seattle WA, October 2023
5. S. M. King, N. S. Sarna, **I. Ortiz**, M. Lopez-Cavestany, Z. Zhang. Invasion of Jurkat Cells Conjugated with E-Selectin Modified Liposomes. *Biomedical Engineering Society 2023 Annual Conference*, Seattle WA, October 2023
6. **I Ortiz**, J.D. Greenlee, P.V. Taufalele, S.C. Schwager, J.A. Mosier, M.R. King, C.A. Reinhart-King. Mesenchymal Phenotype in Migratory Cancer Cell Subpopulations Enhance Fluid Shear Stress Resistance, *Biomedical Engineering Society 2022 Annual Conference*, San Antonio TX, October 2022
7. **I Ortiz**, J.D. Greenlee, P.V. Taufalele, S.C. Schwager, J.A. Mosier, M.R. King, C.A. Reinhart-King. Cytoskeletal Actin Organization of Migratory Cancer Cells Aids Single-Cell Fluid Shear Stress Resistance, *Ford Fellows Conference*, Virtual, October 2022
8. **I Ortiz**, J.D. Greenlee, P.V. Taufalele, S.C. Schwager, J.A. Mosier, M.R. King, C.A. Reinhart-King. Cytoskeletal Actin Organization of Migratory Cancer Cells Aids Single-Cell Fluid Shear Stress Resistance, *VICC 23rd Annual Scientific Retreat*, Nashville TN, May 2022

9. M.R. Zanotelli, **I Ortiz**, J Zhang, W Wang, N.C. Chada, C.A. Reinhart-King. Bioenergetic Changes in Response to Collagen Density Alter Cell Migration Ability, *Biomedical Engineering Society 2021 Annual Conference*, Orlando FL, October 2021
10. **I Ortiz**, S.C. Schwager, P.V. Taufalele, J.A. Mosier, C.A. Reinhart-King. Phenotypic Sorting of Colorectal Cancer Cells Based on Migration To Reveal Heterogeneous EMT, *Biomedical Engineering Society 2021 Annual Conference*, Orlando FL, October 2021
11. **I Ortiz**, S.C. Schwager, J.A. Mosier, P.V Taufalele, C.A. Reinhart-King. Phenotypic Sorting of Migratory Colorectal Cancer Cells Display Phenotypic and Metabolic Differences Among Subpopulations, *Metabolism Immunity & Inflammation of the Mid-South Inaugural Conference*, April 2021
12. **I Ortiz**, S.C. Schwager, J.A. Mosier, C.A. Reinhart-King. Primary Tumor Colorectal Cancer Cells Display Greater Migratory Capabilities than Metastatic Cells in Three-Dimensional Collagen Microtracks, *Nano Day*, Vanderbilt University, Nashville TN, October 2020
13. V Mayo, A Bowles, A Cordoves, L Wubker, **I Ortiz**, D Correa, A Agarwal. Modular Tissue Engineering of Bone Microenvironment as an Extravasation Model, *Biomedical Engineering Society 2020 Annual Conference*, Virtual, October 2020
14. **I Ortiz**, V Mayo, A Agarwal. Tissue Engineered Vascular Wall as an Extravasation Model, *Biomedical Engineering Society 2019 Annual Conference*, Philadelphia PA., October 2019

WORKS ACKNOWLEDGED IN

1. V Mayo. Modular Tissue Engineering of Bone Microenvironment as an Extravasation Model, *University of Miami PhD Dissertation.* December 2020. https://scholarship.miami.edu/esploro/outputs/991031524182702976
2. R Besser. Engineering a Compartmentalized Hydrogel Platform for In Vitro Assembly of Neuromuscular Junctions, *University of Miami PhD Dissertation.* December 2019. https://scholarship.miami.edu/esploro/outputs/991031447529702976
3. M.K. Perera, J.D. Englehardt. Simultaneous Nitrogen and Phosphorous Recovery from Municipal Wastewater by Electrochemical pH Modulation, *Separation and Purification Technology*. May 2020. https://doi.org/10.1016/j.seppur.2020.117166
4. M.K. Perera. Development of the Electrohydromodulation Process for Simultaneous Recovery of Nitrogen and Phosphorus from Municipal Wastewater, *University of Miami PhD Dissertation*. August 2019. https://www.proquest.com/openview/6ec749fc2bf59e59d314ed5fa4d75cf8/1.pdf?pq-origsite=gscholar&cbl=18750&diss=y

MENTORSHIP/LEADERSHIP EXPERIENCE

2023 – 2024 SyBBURE Undergraduate Research Program, Vanderbilt University, Nashville TN
SyBBURE Searle Graduate Fellow
Mentor and train undergraduate students on research techniques in Dr. Cynthia Reinhart-King's lab
Help conceptualize undergraduate student research projects
Design, plan, and oversee experiments performed by undergraduate students

2021 – 2024 Graduate Student Mentor
Undergraduate Mentees: Victor L. Dunagan, Reethi S. Padmanabhan, Mara E. Yella
High School Mentee: Simon King

Camp Kesem Alumni Advisory Board
Advise the Vanderbilt Camp Kesem executive board on ideas and events for Camp Kesem campers, as well as providing logistical support when needed.

BME Underrepresented Minority (URM) Mentorship Program
Graduate student mentor, helping guide undergraduate sophomore, junior and senior URM students through the undergraduate process and share perspective as a fellow URM in STEM.

OUTREACH

2020 – 2024 Elementary Education (EE)

Introduce scientific concepts to elementary school students to enrich their love and understanding of the subject. I particularly create video tutorials of these activities for the students to follow along at home as well as lead them through the activity in person.

Glencliff High School Outreach

Participating in Mr. Adam Guidry's physics and automated systems class giving short lectures on current and cutting-edge scientific topics. Also discussed road for minorities to be successful and attend higher education.

2019 "Independence with Autism", South Florida PBS. Produced by N Bundschu, J Piszczor; Directed & Edited by **I Ortiz** https://www.youtube.com/watch?v=nCU8RRaSiAA&list=LL&index=36

2017 – 2020 Camp Kesem Counselor

Helped raise money for a free one-week sleep-away camp for kids whose parents have or have had cancer. Personally watch and care for campers at camp and guide cabin chats, allowing kids to learn how to express themselves.

Marketing Intern/School Mascot (Sebastian the Ibis)

Represent the University of Miami brand. Organize/travel for marketing/sporting events & commercials/guest appearances. https://miamihurricanes.com/news/2020/04/07/being-sebastian

2017 Volunteer - International Dermatology Research Inc., Supervisor - Ariel F. Sotolongo MD

Shadowed Dr. Sotolongo observing surgical & clinical procedures, especially skin tumor removals & cyst extractions. Gained a better understanding of doctor-patient relations while honing interpersonal skills.

SKILLS

Native in Spanish
2D/3D Cell Culture, Hydrogel Fabrication, Collagen Micro-molding, Cell Spheroid Fabrication
MATLAB, R, Image J, SOLIDWORKS
Scanning Electron Microscopy, Fluorescent Microscopy, Confocal Microscopy
Immunocytochemistry, Western Blotting, Real-Time Quantitative Polymerase Chain Reaction
RNA sequencing, Agilent Seahorse XF Analysis